BRAC UNIVERSITY

Quiz 1 **Department of Computer Science and Engineering**

CSE110: Programming Language I Duration: 40 minutes Marks: 20

Name: (Please write in CAPITAL LETTERS)	ID:	Section: 42

1. Write a python program that will take an integer number of at least three digits from the user and check if the number is a happy number or not. An integer is said to be a happy number if the summation of the last two digits of the number is divisible by 4.
   [Consider the input number to be an INTEGER. You are not allowed to use String indexing for solving this task] [Marks 10]
   [CO 3 + CO 4]

Sample Input 1:

362

Sample output 1:

362 is a Happy number

Explanation:

362 is a happy number because the summation of the last two digits of the number is 6+2=8 which is divisible by 4.

Sample Input 2:

12567

Sample output 2:

12567 is not a Happy number

Explanation:

12567 is not a happy number because the summation of the last two digits of the number is 6+7= 13 which is not divisible by 4.

1. Trace the below table and write the outputs in the given boxes. You MUST show the trace table. [Marks 10] [C0 3 + CO 4]

1	`p, j , k = 0 , 20, 70`
2	`while (k > 5):`
3	`  while ( j < k ):`
4	`    p = k + j -35`
5	`    print(3 + int (p / 2))`
6	`    j += 10`
7	`  k -= 15`
8	`print(True and (False or not True) or not False)`
9	`print(j % 7 + 2)`
10	`print(k > 9 == False)`
11	`print(p == 45)`
12	`print(14/7 + 3)`

Output